

Worksheet # 4

Week 2, Day 2: Obtaining data; populations, frames, and samples; wide vs. long data

Name: _____

NetID: _____

Date: _____

Guided checkpoints (optional)

1. The Cathy O’Neil Quote

Write down the quote about ”opinions.”

2. Target Population, Access Frame, Sample

Draw the three as nested circles. Label each one.

3. Three Ways of Obtaining Data

List the three main ways of obtaining data, in the order you should try them.

4. How Would You Visualize It?

“How does winning a FIFA title affect the birth rate of the winning country?” Sketch one plot (or more) that you would want to see.

5. Wide vs. Long Data

Sketch a dataframe in each format. On each, draw an arrow showing the direction it grows as new data is added.